KEDAR DAMALE

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Portfolio Website · GitHub · LinkedIn

SUMMARY

Applied Data Scientist with 1.5 years of experience building end-to-end AI/ML solutions across fintech, financial automation, pharma, and medical research. Experienced in machine learning, Generative AI, multi-agent systems, and data pipeline development, with a focus on translating research and experimentation into deployable, production-grade applications.

SKILLS

- **Programming and Data** : Python, SQL, NumPy, Pandas, Polars
- **Machine Learning** : Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Generative AI** : LangChain, LangGraph, Multi-Agent Systems, RAG, MCP, Human-in-the-Loop (HITL)
- **Deep Learning** : PyTorch, CNNs, YOLO, Transfer Learning, Model Training and Evaluation
- **Data Engineering and MLOps** : ETL/ELT Pipelines, MLflow, FastAPI, Docker, AWS, CI/CD, Git/GitHub
- **Mathematics and Statistics** : Probability and Statistics, Linear Algebra, Calculus,

WORK EXPERIENCE

ML and GenAI Engineer **Jan 2026 - Present**
Globalspace Technologies Ltd. *Navi Mumbai, MH, India*

Budget Automation and Incentive Calculator

- Replaced an XGBoost model per employee, which became difficult to manage beyond 20-30 people, with a unified hierarchy model designed to scale to thousands of medical representatives (MRs). The solution is under review by large pharmaceutical companies, including Himalaya and Zydus.
- Built a top-down budgeting workflow that sets targets using prior performance and capacity, supports time-bound revision requests, reconciles conflicting requests where possible, and routes proposed changes to managers for approval.
- Developed a DAG-based incentive calculator with a custom AI agent that converts policy PDFs into executable functions at runtime and explains each employee's result, enabling business users to update policies without specialist admin support.

PatGPT

- The earlier workflow relied entirely on retrieval-augmented generation (RAG). It handled simple lookups but could not answer multi-step commercial questions or recommend actions across more than 3 million pharmaceutical sales and tender records.
- Rebuilt PatGPT as an agentic planning assistant for forecasting, simulation, prediction, and historical analysis, with human-review checkpoints and evidence-grounded answers.
- Added streamed responses, safe retry handling, and checksum-verified data promotion and rollback. The system is now used at Zydus.

Freelance Automation Developer **May 2025 - Jan 2026**
Propelligence Advisors Pvt. Ltd. *Thane, MH, India*

- Manual reconciliation of purchase-register invoices against GSTR-2B records took five to six days per cycle; inconsistent vendor, invoice, and date formats made exact-match tools unreliable.
- Engineered a large-file reconciliation pipeline using RapidFuzz with an 80%+ match threshold and date/value tolerances, Polars processing, and constant-memory report generation.
- Processed 10,000+ invoices in 12 seconds, down from five to six days per cycle, reducing processing time by over 99.9% with 100% precision (no false positives). Integrated the module into Granska, an audit suite adopted by two CA firms and about 20 auditors.

PROJECTS

1. Chessablanka: Chess Position Analyzer **GitHub Link**

- No readily available solution could analyze a chess position directly from a board photo, forcing players, including the college IT chess team, to manually transcribe positions into Stockfish during competition preparation.
- Built an OpenCV computer-vision pipeline to detect chessboards and map their grids, training a custom YOLOv8 model on Roboflow with an augmented dataset expanded from 700+ to 1,700 images.
- Delivered a photo-to-analysis tool used by the IT chess team for college competition preparation, achieving 98.57% detection accuracy across 500+ test images and integrating Stockfish for position evaluation and best-move analysis.

2. NeuroTRIBE: Cortical Response Analysis for ADHD Research **GitHub Link**

- Addressed the challenge of reproducibly testing whether ADHD is associated with altered stimulus-evoked cortical responses, where fMRI-to-movie alignment and access-controlled HBN inputs complicate cohort comparisons.
- Built an interactive, Docker Compose-based TRIBE v2/HBN movie-fMRI pipeline aligning observed BOLD signals $y_{v,t}$ with pretrained predictions $\hat{y}_{v,t}$ and comparing confirmed-ADHD with non-ADHD cohorts at vertex, region, and network scales.
- Computed residuals $e_{v,t} = y_{v,t} - \hat{y}_{v,t}$, summarized vertex-wise deviation and correlation, and tested group effects with age-, sex-, site-, and motion-adjusted models plus false-discovery-rate (FDR) correction; rendered findings as 3D cortical maps and reports.

3. Low-Cost Cattle Monitoring with ESP32 and DBSCAN **GitHub Link**

- Conventional cattle trackers raise deployment costs by requiring a GPS sensor on every animal, despite cattle naturally moving in groups.
- Led a team building a lower-cost group-tracking system, replacing per-animal GPS with one GPS-equipped ESP32 master and RSSI-based child nodes to infer relative positions; streamed telemetry to Flask/MongoDB and a live Next.js dashboard.
- Applied DBSCAN to RSSI-informed location points, where $N_\varepsilon(p) = \{q \mid d(p,q) \leq \varepsilon\}$ and a core point satisfies $|N_\varepsilon(p)| \geq$ minPts, to cluster grazing zones and reveal herd movement patterns.

EDUCATION AND ACHIEVEMENTS

- Qualified GATE DA 2026 and graduated in IT Engineering with an 8.5 CGPA from Mumbai University.
- Published an automation suite for CA and audit firms, adopted by two CA firms and used by approximately 20 auditors.